

Construct a grid coordinate system that uses coordinates to locate positions within a space; use coordinates and directional language to describe position and movement

Learning intention

- construct a grid coordinate system that uses coordinates to locate positions within a space; use coordinates and directional language to describe position and movement.

Teach and model

- discussing the conventions of indicating a point in a grid coordinate system.
- placing a coordinate grid over a contour line, drawing and listing the coordinates of each point in the picture.
- investigating how autonomous vehicles use mapping, GPS systems, communication systems and path planning to navigate within a space.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.